```
> restart; #the thermal's flux ; 8 sec
```

```
> eq1 := diff(p(r,t),t$2) + V·diff((p(r,t)·V),r$2) - (V^2·p(r,t))/R = 0;
```

$$eq1 := \frac{\partial^2}{\partial t^2} p(r,t) + V^2 \left(\frac{\partial^2}{\partial r^2} p(r,t) \right) - \frac{V^2 p(r,t)}{R} = 0 \tag{1}$$

```
> eq2 := diff(p(r,t),t) + diff(p(r,t)·V,r) = 0;
```

$$eq2 := \frac{\partial}{\partial t} p(r,t) + \left(\frac{\partial}{\partial r} p(r,t) \right) V = 0 \tag{2}$$

```
> iv1 := p(2·R,0) = 0, D[1](p)(0,0) = -R;
```

$$iv1 := p(2R,0) = 0, D_1(p)(0,0) = -R \tag{3}$$

```
> Sol := pdsolve([eq1, eq2, iv1]);
```

$$Sol := p(r,t) = R^{3/2} \sqrt{2} \sinh\left(\frac{(Vt + 2R - r)\sqrt{2}}{2\sqrt{R}} \right) \operatorname{sech}\left(\sqrt{2}\sqrt{R} \right) \tag{4}$$

```
> Sol
```

$$p(r,t) = R^{3/2} \sqrt{2} \sinh\left(\frac{(Vt + 2R - r)\sqrt{2}}{2\sqrt{R}} \right) \operatorname{sech}\left(\sqrt{2}\sqrt{R} \right) \tag{5}$$

```
> change_of_vars := {r = rb·ρ}
```

$$change_of_vars := \{r = rb\,\rho\} \tag{6}$$

```
> tmp := PDEtools:-dchange(change_of_vars, [eq1, eq2, iv1], [ρ], params = rb);
```

$$tmp := \left[\frac{\partial^2}{\partial t^2} p(\rho,t) + \frac{V^2 \left(\frac{\partial^2}{\partial \rho^2} p(\rho,t) \right)}{rb^2} - \frac{V^2 p(\rho,t)}{R} = 0, \frac{\partial}{\partial t} p(\rho,t) + \frac{\left(\frac{\partial}{\partial \rho} p(\rho,t) \right) V}{rb} = 0, p(2R,0) = 0, D_1(p)(0,0) = -R \right] \tag{7}$$

```
> eq3 := diff(p(ρ,t),t$2) + (V^2·diff(p(ρ,t),ρ$2))/rb^2 - (V^2·p(ρ,t))/R = 0;
```

$$eq3 := \frac{\partial^2}{\partial t^2} p(\rho,t) + \frac{V^2 \left(\frac{\partial^2}{\partial \rho^2} p(\rho,t) \right)}{rb^2} - \frac{V^2 p(\rho,t)}{R} = 0 \tag{8}$$

```
> eq4 := diff(p(ρ,t),t) + (diff(p(ρ,t),ρ)·V)/rb = 0, p(2R,0) = 0;
```

$$eq4 := \frac{\partial}{\partial t} p(\rho, t) + \frac{\left(\frac{\partial}{\partial \rho} p(\rho, t)\right) V}{rb} = 0, p(2R, 0) = 0 \quad (9)$$

> $Sol := pdsolve([eq3, eq4, iv1]);$

$$Sol := p(\rho, t) = \frac{R^{3/2}\sqrt{2}\sinh\left(\frac{\sqrt{2}\left(\left(R - \frac{\rho}{2}\right) rb + \frac{Vt}{2}\right)}{\sqrt{R}}\right)\operatorname{sech}\left(rb\sqrt{2}\sqrt{R}\right)}{rb} \quad (10)$$

> $solve\left(\left\{\left\{\frac{\sqrt{2}R^{3/2}\left(e^{\frac{2\sqrt{2}\left(\left(R-\frac{\rho}{2}\right)rb+\frac{Vt}{2}\right)}{\sqrt{R}}} - 1\right)e^{-\frac{\sqrt{2}(Vt-rb\rho)}{2\sqrt{R}}}}{\left(e^{2rb\sqrt{2}\sqrt{R}}+1\right)rb} = 1, \mathrm{D}[2](p)(1,0) = V, \rho = 1\right\}\right\};$

$$\left\{R = R, V = \mathrm{D}_2(p)(1,0), rb = rb, \rho = 1, t = -\frac{1}{\mathrm{D}_2(p)(1,0)}\left(\sqrt{2}\ln\left(RootOf\left(2_Z^2 R^{3/2} + \left(\sqrt{2}\, rb\, e^{2rb\sqrt{2}\sqrt{R}} + \sqrt{2}\, rb\right)_Z - 2R^{3/2}e^{2rb\sqrt{2}\sqrt{R}}\right)\right)\sqrt{R} - rb\right)\right\} \quad (11)$$

> $restart;$

> $eq1 := \frac{2\cdot m}{h^2} E \cdot f(r,t) + \frac{R}{r} diff((f(r,t)), r\$2) - \frac{1}{a^2 \cdot c^2} diff(f(r,t), t\$2) = 0;$

$$eq1 := \frac{2mEf(r,t)}{h^2} + \frac{R\left(\frac{\partial^2}{\partial r^2} f(r,t)\right)}{r} - \frac{\frac{\partial^2}{\partial t^2} f(r,t)}{a^2 c^2} = 0 \quad (12)$$

> $iv1 := f(2\cdot R, t) = 0, \mathrm{D}[1](f)(0,0) = R, \mathrm{D}[2](g)(rb, 0) = a\cdot c;$

$$iv1 := f(2R, t) = 0, \mathrm{D}_1(f)(0,0) = R, \mathrm{D}_2(g)(rb, 0) = ac \quad (13)$$

> $change_of_vars := \left\{ r = rb\cdot \mathrm{rho}, f(r,t) = \frac{g(\mathrm{rho}, t)\cdot(2\cdot R - rb\cdot \mathrm{rho})}{2\cdot R}\right\};$

$$change_of_vars := \left\{r = rb\,\rho, f(r,t) = \frac{g(\rho,t)\,(-rb\,\rho + 2R)}{2R}\right\} \quad (14)$$

> $tmp := PDEtools:\text{-}dchange(change_of_vars, [eq1, iv1], [g(\mathrm{rho}, t), \mathrm{rho}]);$

$$tmp := \left[\frac{m\,E\,g(\rho,t)\,(-rb\,\rho+2\,R)}{h^2\,R} + \frac{R\left(\frac{\left(\frac{\partial^2}{\partial\rho^2}\,g(\rho,t)\right)(-rb\,\rho+2\,R)}{2\,R} - \frac{\left(\frac{\partial}{\partial\rho}\,g(\rho,t)\right)rb}{R}\right)}{rb^3\,\rho} - \frac{\left(\frac{\partial^2}{\partial t^2}\,g(\rho,t)\right)(-rb\,\rho+2\,R)}{2\,a^2\,c^2\,R} = 0,\ 0 = 0,\ \frac{D_1(g)(0,0) - \frac{g(0,0)\,rb}{2\,R}}{rb} = R,\ D_2(g)(rb,0) = a\,c \right] \tag{15}$$

```
> eq2 := tmp[1];
```

$$eq2 := \frac{m\,E\,g(\rho,t)\,(-rb\,\rho+2\,R)}{h^2\,R} + \frac{R\left(\frac{\left(\frac{\partial^2}{\partial\rho^2}\,g(\rho,t)\right)(-rb\,\rho+2\,R)}{2\,R} - \frac{\left(\frac{\partial}{\partial\rho}\,g(\rho,t)\right)rb}{R}\right)}{rb^3\,\rho} - \frac{\left(\frac{\partial^2}{\partial t^2}\,g(\rho,t)\right)(-rb\,\rho+2\,R)}{2\,a^2\,c^2\,R} = 0 \tag{16}$$

```
> eq3 := tmp[3];
```

$$eq3 := \frac{D_1(g)(0,0) - \frac{g(0,0)\,rb}{2\,R}}{rb} = R \tag{17}$$

```
> eq4 := tmp[4];
```

$$eq4 := D_2(g)(rb,0) = a\,c \tag{18}$$

```
> #eq5:=tmp[5];
```

> $iv2 := \dfrac{D_1(g)(0,0) - \dfrac{g(0,0)\,rb}{2R}}{rb} = R,\ D_2(g)(rb,0) = a\,c;$

$$iv2 := \frac{D_1(g)(0,0) - \frac{g(0,0)\,rb}{2R}}{rb} = R,\ D_2(g)(rb,0) = a\,c \tag{19}$$

> $sys := [eq2, iv2]:\ Sol := pdsolve(sys, HINT = \text{`*`});$

$$Sol := g(\rho,t) = \frac{1}{-rb\,\rho + 2R}\left(\mathrm{AiryBi}\left(-\left(\frac{\left(2Emc^2a^2 - {_c_2}\,h^2\right)rb^3}{Rc^2a^2h^2}\right)^{1/3}\rho\right)\right. \tag{20}$$

$$\left.\left(\frac{ac\left(-rb^2+2R\right)RootOf\left({_Z}^2-{_c_2}\right)\sinh\left(\sqrt{-{_c_2}}\,t\right)}{{_c_2}\,\mathrm{AiryBi}\left(-\left(\frac{\left(2Emc^2a^2 - {_c_2}\,h^2\right)rb^3}{Rc^2a^2h^2}\right)^{1/3}rb\right)}\right.\right.$$

$$\left.\left. - \frac{4\pi\,rb\,3^{1/3}R^2\cosh\left(\sqrt{-{_c_2}}\,t\right)}{3\left(\frac{\left(2Emc^2a^2 - {_c_2}\,h^2\right)rb^3}{Rc^2a^2h^2}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)}\right)\right)$$

> $g(r,t) := -\left(\left(\left(\frac{\left(2Emc^2a^2 - {_c_2}\,h^2\right)rb^3}{Rc^2a^2h^2}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(-\frac{rb^2}{2}+R\right)\cdot\sqrt{-{_c_2}}\right.\right.$

$$\left.\cdot ac\left(e^{\sqrt{-{_c_2}}\,t} - e^{-\sqrt{-{_c_2}}\,t}\right)\right.$$

$$\left. - \frac{1}{3}\left(2\,{_c_2}\,3^{1/3}R^2\pi\,rb\left(e^{\sqrt{-{_c_2}}\,t} + e^{-\sqrt{-{_c_2}}\,t}\right)\mathrm{AiryBi}\left(-\left(\frac{\left(2Emc^2a^2 - {_c_2}\,h^2\right)rb^3}{Rc^2a^2h^2}\right)^{1/3}rb\right)\right)\right)\mathrm{AiryBi}\left(\right.$$

$$-\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\rho\Bigg)\Bigg/\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(rb\,\rho-2\,R\right)_c_2\,\Gamma\left(\frac{2}{3}\right)\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)\right);$$

$$g:=(r,t)\mapsto -\left(\left(\left(\frac{\left(2\cdot E\cdot m\cdot a^2\cdot c^2-_c_2\cdot h^2\right)\cdot rb^3}{R\cdot a^2\cdot c^2\cdot h^2}\right)^{1/3}\cdot\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{rb^2}{2}+R\right)\cdot\sqrt{_c_2}\cdot a\cdot c\right)\cdot\left(e^{\sqrt{_c_2}\cdot t}-e^{-\sqrt{_c_2}\cdot t}\right)\right.$$

$$\left.-\frac{1}{3}\left(2\cdot rb\cdot\pi\cdot 3^{1/3}\cdot R^2\cdot_c_2\cdot\left(e^{\sqrt{_c_2}\cdot t}+e^{-\sqrt{_c_2}\cdot t}\right)\cdot\mathrm{AiryBi}\left(-\left(\frac{\left(2\cdot E\cdot m\cdot a^2\cdot c^2-_c_2\cdot h^2\right)\cdot rb^3}{R\cdot a^2\cdot c^2\cdot h^2}\right)^{1/3}\cdot rb\right)\right)\cdot\mathrm{AiryBi}\left(-\left(\frac{\left(2\cdot E\cdot m\cdot a^2\cdot c^2-_c_2\cdot h^2\right)\cdot rb^3}{R\cdot a^2\cdot c^2\cdot h^2}\right)^{1/3}\cdot\rho\right)\right)\Bigg/\left(\left(\frac{\left(2\cdot E\cdot m\cdot a^2\cdot c^2-_c_2\cdot h^2\right)\cdot rb^3}{R\cdot a^2\cdot c^2\cdot h^2}\right)^{1/3}\cdot(rb\cdot\rho-2\cdot R)\cdot_c_2\cdot\Gamma\left(\frac{2}{3}\right)\cdot\mathrm{AiryBi}\left(-\left(\frac{\left(2\cdot E\cdot m\cdot a^2\cdot c^2-_c_2\cdot h^2\right)\cdot rb^3}{R\cdot a^2\cdot c^2\cdot h^2}\right)^{1/3}\cdot rb\right)\right) \tag{21}$$

> $G:=subs\left(\sqrt{_c_2}=C1,\ _c_2=C1^2,\ \Gamma\left(\frac{2}{3}\right)=1.354117939,\ g(\rho,t)\right);$

$$G:=-\left(0.7384881119\left(1.354117939\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(-\frac{rb^2}{2}+R\right)C1\,a\,c\left(e^{C1\,t}-e^{-C1\,t}\right)\right.\right. \tag{22}$$

$$-\frac{2\,CI^2\,3^{1/3}\,R^2\,\pi\,rb\left(e^{CI\,t}+e^{-CI\,t}\right)\,\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)}{3}\Bigg)$$

$$\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\rho\right)\Bigg)\Bigg/$$

$$\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(rb\,\rho-2\,R\right)CI^2\,\mathrm{AiryBi}\left(\right.\right.$$

$$\left.\left.-\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)\right)$$

> $$G1 := subs\left(\Gamma\left(\frac{2}{3}\right)=1.354117939,-\left(\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(-\frac{rb^2}{2}\right.\right.\right.\right.$$

$$\left.\left.\left.\left.+R\right)CI\,a\,c\left(e^{CI\,t}-e^{-CI\,t}\right)\right.\right.\right.$$

$$\left.\left.-\frac{2\,CI^2\,3^{1/3}\,R^2\,\pi\,rb\left(e^{CI\,t}+e^{-CI\,t}\right)\,\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)}{3}\right)$$

$$\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\rho\right)\Bigg)\Bigg/$$

$$\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(rb\,\rho-2\,R\right)CI^2\,\Gamma\left(\frac{2}{3}\right)\mathrm{AiryBi}\left(\right.\right.$$

$$\left.\left.-\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)\right)\Bigg);$$

$$G1 := -\left(0.7384881119\left(1.354117939\left(\frac{\left(2\,E\,m\,a^2\,c^2-CI^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(-\frac{rb^2}{2}\right.\right.\right. \tag{23}$$

$$+ R \Bigg) C1\, a\, c \left(e^{C1\, t} - e^{-C1\, t} \right)$$

$$- \frac{2\, C1^2\, 3^{1/3}\, R^2\, \pi\, rb \left(e^{C1\, t} + e^{-C1\, t} \right) \mathrm{AiryBi}\left(- \left(\frac{\left(2\, E\, m\, a^2\, c^2 - C1^2\, h^2 \right) rb^3}{R\, a^2\, c^2\, h^2} \right)^{1/3} rb \right)}{3} \Bigg)$$

$$\mathrm{AiryBi}\left(- \left(\frac{\left(2\, E\, m\, a^2\, c^2 - C1^2\, h^2 \right) rb^3}{R\, a^2\, c^2\, h^2} \right)^{1/3} \rho \right) \Bigg) \Bigg/$$

$$\left(\left(\frac{\left(2\, E\, m\, a^2\, c^2 - C1^2\, h^2 \right) rb^3}{R\, a^2\, c^2\, h^2} \right)^{1/3} \left(rb\, \rho - 2\, R \right) C1^2\, \mathrm{AiryBi}\left(- \left(\frac{\left(2\, E\, m\, a^2\, c^2 - C1^2\, h^2 \right) rb^3}{R\, a^2\, c^2\, h^2} \right)^{1/3} rb \right) \right)$$

> $m := 9.109383560\ 10^{-31}$; $\epsilon := 8.854187815\ 10^{-12}$; $h := \frac{6.626070040\ 10^{-34}}{2 \cdot \mathrm{Pi}}$; $e := 1.602176620\ 10^{-19}$; $c := 299792458$; $rb := 5.291772105\ 10^{-11}$; $Tq := 1.76536 \cdot 10^{-19}$; $E := 2.179 \cdot 10^{-18}$; $a := 0.007297352566$; $R := 3.9228 \cdot 10^{22}$; $ms := -9.284764612\ 10^{-24}$; $hm := \frac{3.151228184\ 10^{-27}}{2 \cdot \mathrm{Pi}}$; $Sq := 4.3 \cdot 10^{-19}$; $C1 := 1$;

$$m := 9.109383560 \times 10^{-31}$$

$$\epsilon := 8.854187815 \times 10^{-12}$$

$$h := 1.054571800 \times 10^{-34}$$

$$e := 1.602176620 \times 10^{-19}$$

$$c := 299792458$$

$$rb := 5.291772105 \times 10^{-11}$$

$$Tq := 1.765360000 \times 10^{-19}$$

$$E := 2.179000000 \times 10^{-18}$$

$$a := 0.007297352566$$

$$R := 3.922800000 \times 10^{22}$$

$$ms := -9.284764612 \times 10^{-24}$$

$$hm := 5.015335422 \times 10^{-28}$$

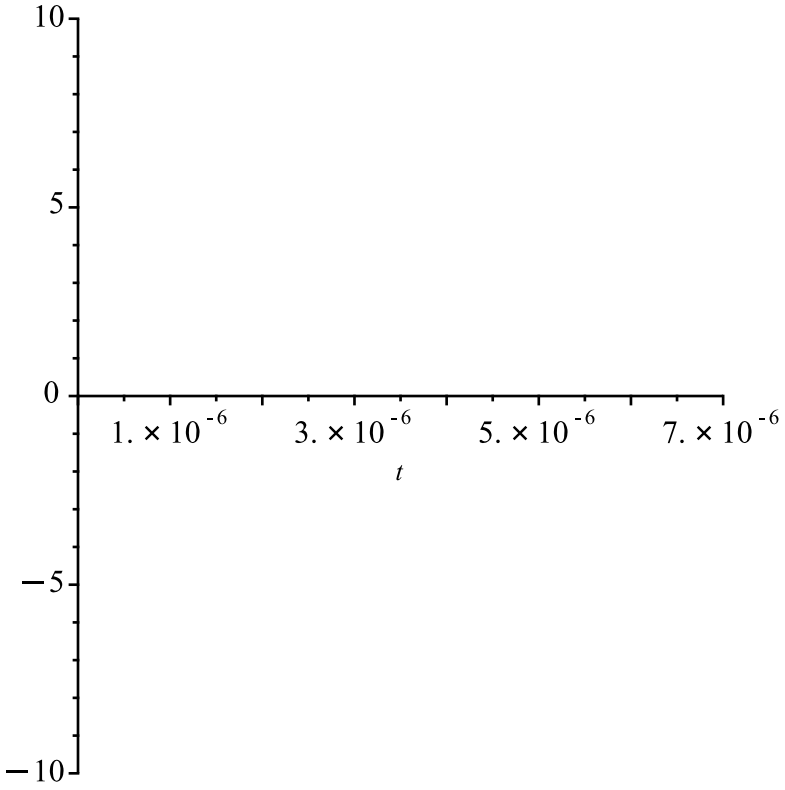
$$Sq := 4.300000000 \times 10^{-19}$$

$$C1 := 1 \quad \textbf{(24)}$$

> $\#\rho := 10^{-19}$; $evalf\left(\Gamma\left(\frac{2}{3}\right)\right)$;$\#\rho$ *constant*;

> $plot\left(G1, t = 0..7\cdot 10^{-6}\right)$;

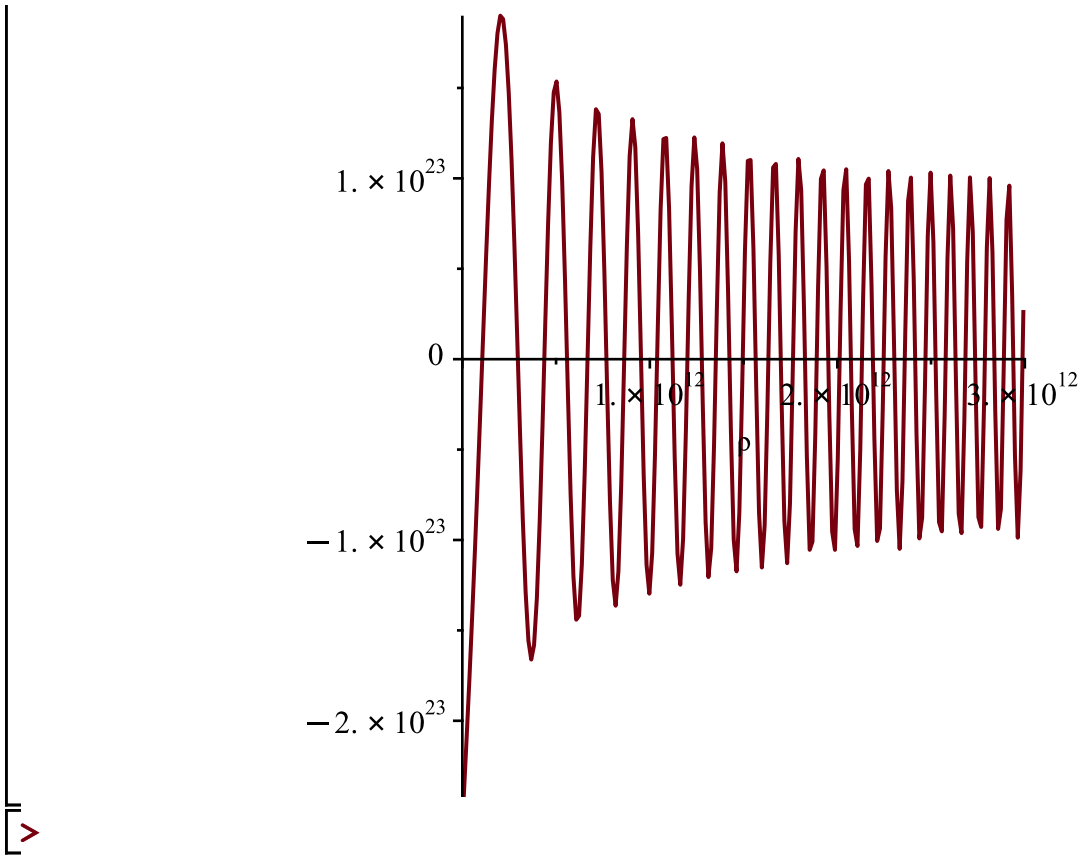
Warning, expecting only range variable t in expression -.1087034087e12*(.1283854290e19*exp(t)-.1283854290e19*exp(-t) -.1048758222e36*3^(1/3)*(exp(t)+exp(-t)))*AiryBi(-.1104783161e-10*rho)/(.5291772105e-10*rho-.7845600000e23) to be plotted but found name rho



> $t := 0$;$\#t$ *constant; Wave function*

$$t := 0 \quad \textbf{(25)}$$

> $plot\left(G, \rho = 0..3\cdot 10^{12}\right)$;


1. × 10^23
0
−1. × 10^23
−2. × 10^23
1. × 10^12
2. × 10^12
3. × 10^12
p
V